



Montana State takes a stand

Montana State says that it will not do business with any ISPs that violate Net Neutrality
<http://digit.in/montana4nn>



Windows that generate electricity

Research groups report that they have created perovskite-tinted windows that can harvest solar energy. <http://digit.in/solarwindos>

MSI GE73VR 7RF Gaming Laptop

Fancy a GTX 1070 with a 120Hz panel?

Gaming laptops with the high-end GPUs were generally really massive and heavy. The MSI GE73VR 7RF offers a GTX 1070 with most of the bulk cut out. It's not a MAX-Q laptop so you're getting a powerful GTX 1070 along with a 17.3-inch 120 Hz panel. The competition in the high-end gaming laptop segment is heating up since most of the mainstream laptop manufacturers have also launched their gaming lineups in India. So let's see how the MSI GE73VR 7RF ranks among them.

Build Quality

Being a bigger laptop, the MSI GE73VR 7RF uses plenty of metal to for rigidity. Most laptops tend to use plastic for the display panel backing. That's not the case here. However, the inside seems to have more plastic than we like. The unit is well cooled, even at peak loads, however, there are exhausts placed towards the sides which makes using a mouse on either side a bit uncomfortable. Sure, it's winter right now and we're liking the exhaust keeping our hands warm but India is hot for the majority of the year. So this arrangement isn't appealing for the rest of the year. A rear-facing exhaust would have been optimal.

Speaking of cooling, we see dedicated heat pipes for the CPU and the GPU. And even the SSD has a nice metal heatsink. The Toshiba HG6 isn't known to be all that hot, but if we were to swap it out for an NVMe SSD, then we're glad to have a heatsink rather than not have it but need it. The laptop weighs about 2.88 kg on its own and the 230 W adapter isn't light by any means either. However, this is a proper gaming laptop so we're going to look the other way.



Price
₹1,85,990



Performance

Let's break down the performance as per each component.

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Storage

The Toshiba HG6 SSD isn't a great performer. It's peak speeds are about 540-550 MBps which is what SATA SSDs are capped at. Since NVMe SSDs have become really cheap these day, we'd have much preferred to see a 256 GB NVMe drive. It won't be enough to have a lot of games installed but intensive software like Photoshop and Premiere Pro will certainly benefit a lot.

SD Card Reader

For the SD card reader, we use a 32 GB Lexar 1000x UHS-II SD card which easily gives 135 MB/s read speeds and 55 MB/s write speeds on USB 3.0 SD card readers and on the MSI GE73VR 7RF we managed to get about 43 MB/s. Like most gaming laptops we've tested, this isn't something manufacturers focus much on and most of the times, it is a simple Realtek solution.

Wi-Fi

For Wi-Fi and Bluetooth, the MSI GE73VR 7RF makes use of yet another Killer network controller - the 1435. Killer's solutions provide built in QoS which makes a difference when you

are gaming while running tasks in the background. However, when you are looking at performance in everyday tasks then there isn't a difference worth pointing out.

Gaming

A combination of the Intel Core i7 7700HQ with the NVIDIA GTX 1070 is quite powerful for gaming. We're left with a 1080p screen but you switch on DSR for a much higher visual fidelity. We managed to get 17823 in Fire Strike, 8496 in Fire Strike Extreme and 4182 in Fire Strike Ultra.

In TimeSpy, we scored 5388 and in Time Spy Extreme we got 2481 which are all very much close with the other GTX 1070 machines we've tested.

And now for the game benchmarks. In Rise of the Tomb Raider, we scored 92 FPS on Ultra while The Witcher 3 came slightly short of 60 FPS. Bioshock Infinite being an older game scored about 136 FPS. It goes without saying that the GTX 1070 on the MSI GE73VR 7RF makes high-end gaming quite easy.

Verdict

The MSI GE73VR 7RF is great for gaming with its GTX 1070 and it packs a wonderful 120 Hz display as well. However, we'd have preferred a better SSD on the laptop given what it has to compete against. Nevertheless, if you want a beefy laptop to play the latest AAA titles without breaking a sweat, then the GE73VR 7RF is a great contender.

—Mithun Mohandas

SPECIFICATIONS

PROCESSOR: Intel Core i7 7700HQ | PLATFORM: HM175 | RAM: 16 GB 2400 MHz | SCREEN SIZE: 17.3-inches 120Hz | SCREEN RESOLUTION: 1920x1080 | GPU: GTX 1070 8 GB GDDR5X | SSD: Toshiba THNSNJ256G8NY 256 GB | BATTERY: 51 Whr | AC ADAPTER: 230 W | WEIGHT: 2.88 Kg

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